

Read-modify-write (RMW) operations:

exchange(x): store x and return old value

compare_exchange_strong(expected, desired)

compare_exchange_weak(expected, desired): allowed to fail spuriously, i.e. behave as if $\text{expected} \neq \text{desired}$ even when $\text{expected} == \text{desired}$. Failure should occur with low probability.

Compare_Exchange: Consider: $x.\text{compare_exchange}(e, d)$;

The value of x and e are compared.

- If equal: d is stored to x and true is returned.
- If not equal: x is stored to e, x is left unchanged and false is returned.

Comparing and exchanging combined into one indivisible operation (RMW). If two threads try to concurrently compare-exchange x from a to b ($a \neq b$), at most one will succeed.